VIT®

Vellore Institute of Technology

(Deemed to be University under section 3 of UGC Act, 1956)

SCHOOL OF COMPUTER SCIENCE AND ENGINEERING

B.Tech, Winter Semester, 2025-26

Course Code	: BCSE301L	**Deadline**	**: 13.03.2026**
Course Name	: **Software Engineering**	**Max. Marks**	**: 10**
Faculty-In-Charge:	Dr. S. YOGANAND	**Slot**	**:** A1+TA

CASE STUDY

Case Scenario

A startup company has developed an **YOUR PROJECT** that includes the following components:

Component	**Description**
Mobile Application	Android/iOS app for customers to browse restaurants, place orders, and track delivery
Web Application	Admin dashboard for restaurant management, order tracking, and analytics
Backend System	REST API services handling orders, payments, and notifications
DevOps Pipeline	CI/CD pipeline for automated build, testing, and deployment
Database	Stores user details, menu items, and order history
Object-Oriented Design	System developed using OOP concepts such as classes, inheritance, and polymorphism

During testing, the development team wants to ensure **high reliability and quality** before the system is deployed to production. The QA team decides to apply different testing techniques including **Mutation Testing, UI Testing, Mobile App Testing, Web Testing, DevOps Testing, and Object-Oriented Testing (OOT)**.

Q.No	**Question**
1	Identify and explain how **Mutation Testing** can be applied to test the order calculation module (e.g., price, discount, tax). Provide an example of a mutation operator.
2	Describe the **UI testing strategy** that should be used to ensure usability and consistency across the mobile and web interfaces.

3	Explain how **Mobile App Testing** can be conducted to verify compatibility, performance, and functionality across different devices and operating systems.
4	Propose a **Web Application Testing plan** to verify the functionality of the admin dashboard including login, order tracking, and report generation.
5	The project uses a **DevOps CI/CD pipeline**. Explain how automated testing can be integrated into the pipeline to improve software quality.
6	The system follows **Object-Oriented Programming principles**. Explain how **Object-Oriented Testing (OOT)** can be applied to test classes such as User, Order, and Payment.

OUTCOME

Outcome	**Description**
Apply testing techniques	Understand mutation, UI, mobile, and web testing
Analyze DevOps testing	Explain automated testing in CI/CD pipelines
Apply OOT testing	Test object-oriented software components
Evaluate software quality	Identify testing strategies for complex systems

NOTE:

1. Everyone must upload the document **in VTOP on or before the deadlines**.
2. All the testing must be done using any of the tools
3. Rubrics for this assignment is all the contents with **neat test case score full marks** and for **no tool usage 5 marks** will be deducted and if any copy paste found in document only 4 marks will be awarded for whole document.
4. Late submission **will be deducted 2 marks**.